

# PSTAT 127 HW 5

Evan Jamison

2025-11-25

1.

a.

```
library(faraway)
data(wbca)
names(wbca)
```

```
## [1] "Class" "Adhes" "BNucl" "Chrom" "Epith" "Mitos" "NNucl" "Thick" "UShap"
## [10] "USize"
```

```
model <- glm(Class ~ ., data = wbca, family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = Class ~ ., family = binomial, data = wbca)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) 11.16678    1.41491   7.892 2.97e-15 ***
## Adhes       -0.39681    0.13384  -2.965  0.00303 **
## BNucl       -0.41478    0.10230  -4.055  5.02e-05 ***
## Chrom      -0.56456    0.18728  -3.014  0.00257 **
## Epith      -0.06440    0.16595  -0.388  0.69795
## Mitos      -0.65713    0.36764  -1.787  0.07387 .
## NNucl      -0.28659    0.12620  -2.271  0.02315 *
## Thick      -0.62675    0.15890  -3.944  8.01e-05 ***
## UShap      -0.28011    0.25235  -1.110  0.26699
## USize       0.05718    0.23271   0.246  0.80589
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 881.388  on 680  degrees of freedom
## Residual deviance:  89.464  on 671  degrees of freedom
## AIC: 109.46
##
## Number of Fisher Scoring iterations: 8
```

```
model$deviance
```

```
## [1] 89.4642
```

```
model$df.residual
```

```
## [1] 671
```

b.

```
full_model <- glm(Class ~ ., data = wbca, family = binomial)
```

```
reduced_modeling <- step(full_model, direction = "backward")
```

```
## Start:  AIC=109.46
## Class ~ Adhes + BNucl + Chrom + Epith + Mitos + NNucl + Thick +
##         UShap + USize
##
##           Df Deviance    AIC
## - USize   1    89.523 107.52
## - Epith   1    89.613 107.61
## - UShap   1    90.627 108.63
## <none>      89.464 109.46
## - Mitos   1    93.551 111.55
## - NNucl   1    95.204 113.20
## - Adhes   1    98.844 116.84
## - Chrom   1    99.841 117.84
## - BNucl   1   109.000 127.00
## - Thick   1   110.239 128.24
##
## Step:  AIC=107.52
## Class ~ Adhes + BNucl + Chrom + Epith + Mitos + NNucl + Thick +
##         UShap
##
##           Df Deviance    AIC
## - Epith   1    89.662 105.66
## - UShap   1    91.355 107.36
## <none>      89.523 107.52
## - Mitos   1    93.552 109.55
## - NNucl   1    95.231 111.23
## - Adhes   1    99.042 115.04
## - Chrom   1   100.153 116.15
## - BNucl   1   109.064 125.06
## - Thick   1   110.465 126.47
##
## Step:  AIC=105.66
## Class ~ Adhes + BNucl + Chrom + Mitos + NNucl + Thick + UShap
##
##           Df Deviance    AIC
## <none>      89.662 105.66
## - UShap   1    91.884 105.88
```

```
## - Mitos 1 93.714 107.71
## - NNucl 1 95.853 109.85
## - Adhes 1 100.126 114.13
## - Chrom 1 100.844 114.84
## - BNucl 1 109.762 123.76
## - Thick 1 110.632 124.63
```

```
summary(reduced_modeling)
```

```
##
## Call:
## glm(formula = Class ~ Adhes + BNucl + Chrom + Mitos + NNucl +
##      Thick + UShap, family = binomial, data = wbca)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  11.0333      1.3632   8.094 5.79e-16 ***
## Adhes        -0.3984      0.1294  -3.080 0.00207 **
## BNucl        -0.4192      0.1020  -4.111 3.93e-05 ***
## Chrom        -0.5679      0.1840  -3.085 0.00203 **
## Mitos        -0.6456      0.3634  -1.777 0.07561 .
## NNucl        -0.2915      0.1236  -2.358 0.01837 *
## Thick        -0.6216      0.1579  -3.937 8.27e-05 ***
## UShap        -0.2541      0.1785  -1.423 0.15461
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 881.388  on 680  degrees of freedom
## Residual deviance:  89.662  on 673  degrees of freedom
## AIC: 105.66
##
## Number of Fisher Scoring iterations: 8
```

c.

```
names(wbca)
```

```
## [1] "Class" "Adhes" "BNucl" "Chrom" "Epith" "Mitos" "NNucl" "Thick" "UShap"
## [10] "USize"
```

```
newpatient <- data.frame(
  Adhes = 1, BNucl = 1, Chrom = 3,
  Epith = 2, Mitos = 1, NNucl = 1,
  Thick = 4, UShap = 1, USize = 1
)
predict(reduced_modeling, newdata = newpatient, type = "response")
```

```
##          1
## 0.9921115
```

d.

```
p_hat <- predict(reduced_modeling, type = "response")
prediction_class <- ifelse(p_hat > 0.5, "benign", "malignant")
table(prediction_class, wbca$Class)
```

```
##
## prediction_class    0    1
##      benign      11 434
##      malignant 227    9
```

```
type_1 <- sum(prediction_class == "malignant" & wbca$Class == 1)
type_2 <- sum(prediction_class == "benign" & wbca$Class == 0)
type_1
```

```
## [1] 9
```

```
type_2
```

```
## [1] 11
```

2.

a.

```
library(datasets)
library(glmnet)
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-8
```

```
carsCols = as.matrix(mtcars[,c(3,4,5,6,7)])
```

```
head(carsCols)
```

```
##           disp  hp drat   wt  qsec
## Mazda RX4      160 110 3.90 2.620 16.46
## Mazda RX4 Wag  160 110 3.90 2.875 17.02
## Datsun 710      108  93 3.85 2.320 18.61
## Hornet 4 Drive  258 110 3.08 3.215 19.44
## Hornet Sportabout 360 175 3.15 3.440 17.02
## Valiant        225 105 2.76 3.460 20.22
```

```
tail(carsCols, 4)
```

```
##           disp  hp drat   wt  qsec
## Ford Pantera L 351 264 4.22 3.17 14.5
## Ferrari Dino   145 175 3.62 2.77 15.5
## Maserati Bora   301 335 3.54 3.57 14.6
## Volvo 142E     121 109 4.11 2.78 18.6
```

```
dim(carsCols)
```

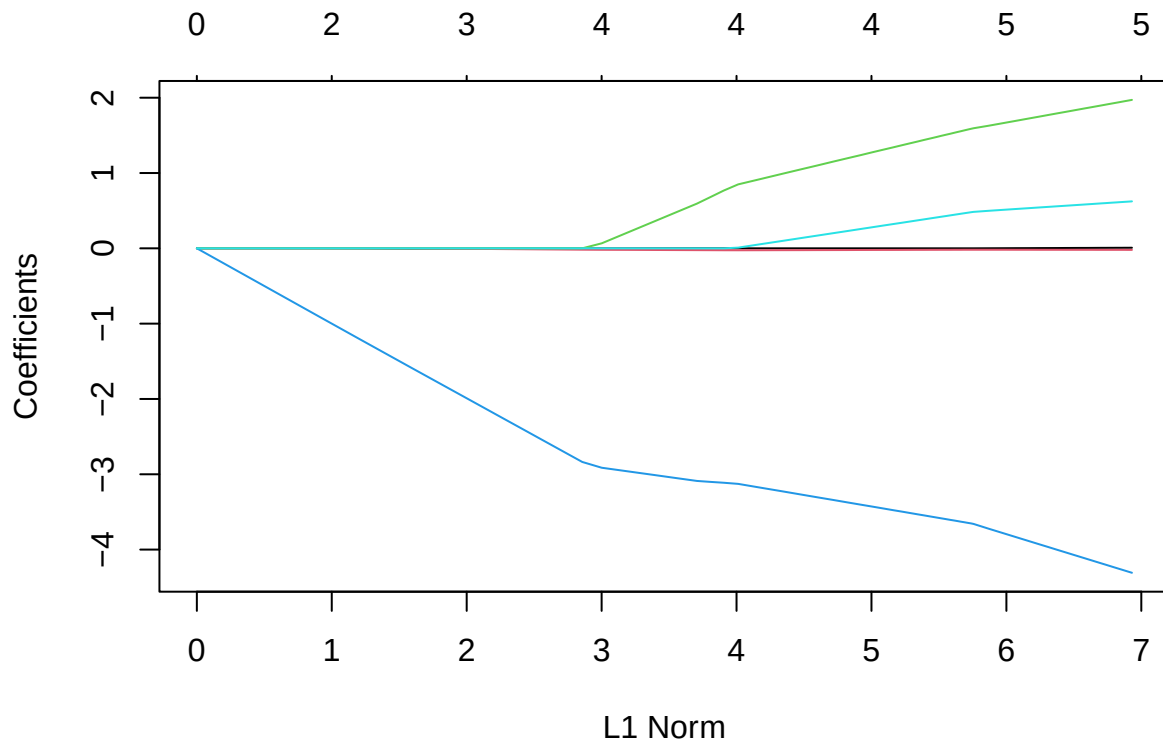
```
## [1] 32 5
```

```
fit.glmnet1 = glmnet( x= carsCols, y= mtcars[,1], alpha=1)
```

(part b I did by hand on a seperate upload)

c.

```
plot(fit.glmnet1)
```



The plot shows lasso coefficient paths for each predictors as lambda penalty varies. On the far left of the graph all coefficients are at 0. As the L1 Norm values increase and we move farther right along the graph, the dark blue line denoting a coefficient becomes negative. This variable is likely strongly associated with the response. As lambda continues to decrease, more lines begin to deviate from 0. Green and cyan in particular begin to increase. At the far right as lambda continues to decrease significantly most predictors are non-zero, and the L1 Norm is the largest. Therefore overall, as the graph and L1 Norm values increase and move right, the coefficients become stronger and more significant in explaining the response values.

(part d is done by hand on a seperate upload)

e.

```
fit.glmnet.5foldCV = cv.glmnet(x=carsCols, y = mtcars[,1], alpha = 1, nfolds = 5)
coef(fit.glmnet.5foldCV, s = "lambda.min")
```

```
## 6 x 1 sparse Matrix of class "dgCMatrix"
##              s1
## (Intercept) 21.22939509
## disp        .
## hp          -0.01926068
## drat         1.52169763
## wt          -3.60548549
## qsec         0.43775607
```

write up done by hand on separate upload